Project Design Phase
Proposed Solution

Date	15 February 2026
Team ID	LTVIP2026TMIDS80513
Project Name	TransLingua: AI-Powered Multi-Language Translator
Maximum Marks	2 Marks

Proposed Solution :

Project team shall fill the following information in the proposed solution table.

S.No.	Parameter	Description
1.	**Problem Statement**	Traditional translation tools often lack contextual awareness, resulting in "robotic" outputs that fail to preserve professional tone for businesses, technical accuracy for researchers, and real-time accessibility for travelers. Users face "navigational anxiety" and professional frustration when specialized jargon or brand voices are lost in translation.
2.	**Idea / Solution Description**	**TransLingua** is a cutting-edge web application built with **Streamlit** and **Google's Generative AI (Gemini)**. It provides a seamless interface for translating text and documents across multiple languages. Key features include context-aware toggles (Professional, Academic, Travel) to ensure the AI adapts its tone and vocabulary to the specific user scenario.
3.	**Novelty / Uniqueness**	Unlike generic translators, TransLingua utilizes **contextual intelligence** to preserve complex academic citations and business brand identities. It features a **Scenario-Based UI** that adapts its layout,such as a high-contrast mobile interface for travelers or a split-screen view for researchers,to optimize the user experience for specific tasks.
4.	**Social Impact / Customer Satisfaction**	The solution enhances global collaboration by breaking language barriers for academics and enables businesses to expand into non-English speaking regions with confidence. It reduces traveler anxiety by providing instant, reliable translations for essential navigation like street signs and menus.
5.	**Business Model (Revenue Model)**	The project follows a **Freemium Model**: basic text translation is free, while a subscription-based "Pro" tier offers advanced features like document file uploads (PDF/LaTeX), unlimited context-aware professional modes, and API access for enterprise localization.

6.	**Scalability of the Solution**	The architecture is highly scalable, utilizing a **3-tier micro-services approach** hosted on cloud platforms like GCP or AWS. This allows the system to handle increasing concurrent users and largescale document processing by leveraging distributed cloud servers and load balancers.